TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Radius of Curvature of a Lens by Newton’s Rings

Interference in the air film between lens and plate draws concentric rings whose diameters, plotted and differenced, give R = (D²_{m+p} − D²_m)/(4pλ) — a metre-scale radius from millimetre rings.

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1. Aim and theory

Aim: to find the radius of curvature R of a plano-convex lens’s curved surface. The lens rests, curved side down, on a flat glass plate, trapping an air film that is thinnest (zero) at the contact point and thickens outward. Monochromatic light reflected from the film’s two surfaces interferes: the path difference is 2t (plus a half-wave change at the plate’s surface), so dark rings form where 2t = mλ. The film thickness at radius r from the contact is t ≈ r²/(2R), giving the dark-ring condition

$$D_m^2 = 4\, m\, \lambda\, R \quad (D_m = \text{diameter of the m-th dark ring})$$

In practice R is taken from the **difference** of two rings’ squared diameters, R = (D²_{m+p} − D²_m)/(4pλ), which cancels the effect of the contact not being a perfect point (dust slightly flattens the centre in every real apparatus).

2. Apparatus and procedure

- **Apparatus:** plano-convex lens on an optical glass plate, glass plate at 45° to reflect light downward, sodium lamp (λ = 589.3 nm), travelling microscope.
- Clean lens and plate meticulously (dust shows as distorted rings). Form the rings and view them through the microscope focused on the film.
- Move the microscope across a diameter: set the cross-wire tangentially on successive dark rings (say the 5th to the 20th) on the left, pass the centre, and continue to the same rings on the right — always moving the microscope in one direction to avoid backlash. Diameters are the right-minus-left reading pairs.
- Plot D² against ring number m: a straight line of slope 4λR. Compute R from the slope or from ring pairs p apart.

3. Worked calculation (model data)

RESULT

Measured dark-ring diameters: D_{10} = 4.86 mm, D_{20} = 6.87 mm. λ = 589.3 nm = 5.893 × 10^{-7} m.
$D_{20}^2 - D_{10}^2 = (6.87^2 - 4.86^2)$ mm² = (47.20 − 23.62) = 23.58 mm² = 2.358 × 10^{-5} m².
R = (D²_{m+p} − D²_m)/(4pλ) with p = 10: = 2.358 × 10^{-5}/(40 × 5.893 × 10^{-7}) = 2.358 × 10^{-5}/2.357 × 10^{-5} = **1.00 m**.
A surface so gently curved that no caliper could feel it is measured to three figures by counting rings — the point of the experiment.

4. Precautions

- Approach every ring setting with the microscope moving in the same direction (screw backlash otherwise shifts diameters).
- Set the cross-wire on the middle of each dark ring's thickness, tangentially, at the same side convention throughout.
- Dust or a hair at the contact makes the central rings elliptical and the centre offset — clean and re-seat the lens; use ring pairs away from the centre (the differencing method forgives a slightly imperfect contact).
- The lens must not be pressed or screwed down hard: strain in the glass distorts the film's geometry.
- Use rings of the same order on both sides (left 10th with right 10th) — misidentifying a ring by one spoils its diameter completely.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is the centre of the pattern dark in reflected light?
2. With λ = 589.3 nm, D_5 = 3.43 mm and D_{15} = 5.95 mm, compute R.
3. Why is R computed from a difference of squared diameters rather than from one ring alone?
4. The liquid of refractive index μ is introduced between lens and plate. How do the ring diameters change, and how does that turn this apparatus into a refractometer?
5. Why do the rings get closer together as the order m increases?
6. State two faults that make the rings non-circular, and their remedies.

Answer key

1. At the contact point the film thickness is zero, so the geometric path difference is zero — but the ray reflected from the lower surface (glass plate, optically denser medium) suffers a phase change of π, equivalent to λ/2. The two reflections are therefore exactly out of phase at the centre: destructive interference, a dark spot.
2. $D_{15}^2 - D_5^2 = (5.95^2 - 3.43^2) = (35.40 - 11.76) = 23.64$ mm² $= 2.364 \times 10^{-5}$ m². $R = 2.364 \times 10^{-5}/(4 \times 10 \times 5.893 \times 10^{-7}) = 2.364 \times 10^{-5}/2.357 \times 10^{-5} = 1.003$ m ≈ 1.00 m.
3. A real contact is never a mathematical point — dust or slight flattening makes the effective zero of thickness uncertain, shifting every D^2 by the same additive constant. Subtracting two rings' D^2 cancels that constant; a single ring would carry the full error into R.
4. The film's optical thickness becomes μt, so the dark-ring condition becomes $D_m^2 = 4m\lambda R/\mu$: all diameters shrink by $1/\sqrt{\mu}$. Measuring the shrinkage of a chosen ring (or the D^2 slope) before and after introducing the liquid gives μ directly — Newton's rings as a refractometer.
5. Because D^2 grows in equal steps ($D^2 \propto m$) while D itself grows only as √m: equal increments of D^2 mean ever-smaller increments of D. The rings therefore crowd together outward — and the crowding is itself a visual check of the $D^2 \propto m$ law.
6. (i) Dust or a fibre trapped at the contact — clean both surfaces and re-seat the lens. (ii) A strained or unevenly pressed lens, or a plate that is not optically flat — release any clamping pressure; use a certified flat plate. (A tilted illumination plate gives uneven brightness rather than non-circular rings — a separate fault with a separate fix.)
